

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0347

Hanging Rock 230 kV
October 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0347. The Interconnection Customer proposes a 88.7 MW solar photovoltaic facility interconnecting at the Hanging Rock 230 kV substation in the Ohio area. The request was received on June 24, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 201 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$766,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Manatee - Hinton Ap 345 KV, the Hinton Ap 345/138kV Autotransformer loads to 130.8% of its 346 MVA rating. Project DFAX: 5.1%.

For the loss of Pt Arena - Weedpatch 500 KV line, the W Canton Substation 500/230 kV TX loads to 101.1% of its 1,168 MVA rating. Project DFAX: 6.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hinton Ap 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$14,468,000	2% (\$262,000)
W Canton Substation 500/230 kV TX	Replace transformer with higher-rated unit	\$27,970,000	2% (\$504,000)
	Total allocated to project		\$766,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.